

Volumes by the Disc and Washer Methods

ANSWER KEY



The disc method (about the x-axis)

- 1 Find the volume of the solid formed by revolving $y = \sqrt{x}$ about the x-axis on $[0, 4]$.

$$V = \pi \int_0^4 (\sqrt{x})^2 dx = \pi \int_0^4 x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = \pi(8) = 8\pi.$$

- 2 Find the volume of the solid formed by revolving $y = x^2$ about the x-axis on $[0, 2]$.

$$V = \pi \int_0^2 (x^2)^2 dx = \pi \int_0^2 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^2 = \pi \cdot \frac{32}{5} = \frac{32\pi}{5}.$$

The disc method (about the y-axis)

- 3 Find the volume formed by revolving $x = y^2$ about the y-axis on $y \in [0, 2]$.

$$V = \pi \int_0^2 (y^2)^2 dy = \pi \int_0^2 y^4 dy = \pi \left[\frac{y^5}{5} \right]_0^2 = \frac{32\pi}{5}.$$

The washer method

- 4 Find the volume formed by revolving the region between $y = x$ and $y = x^2$ (on $[0, 1]$) about the x-axis.

Outer radius $R = x$ (larger on $[0, 1]$), inner radius $r = x^2$.

$$V = \pi \int_0^1 [x^2 - (x^2)^2] dx = \pi \int_0^1 (x^2 - x^4) dx = \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}.$$

- 5 Find the volume formed by revolving the region between $y = \sqrt{x}$ and $y = x$ about the x-axis on $[0, 1]$.

On $[0, 1]$, $\sqrt{x} > x$, so outer radius $R = \sqrt{x}$, inner radius $r = x$.

$$V = \pi \int_0^1 [x - x^2] dx = \pi \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \pi \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{\pi}{6}.$$

Volumes about a horizontal or vertical line (not an axis)

- 6 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 4$ about the line $y = 4$.

Intersections at $x = \pm 2$. Radius = $4 - x^2$ (distance from curve to the line $y = 4$).

$$V = \pi \int_{-2}^2 (4 - x^2)^2 dx = \pi \int_{-2}^2 (16 - 8x^2 + x^4) dx = \pi \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_{-2}^2 = 2\pi \left(32 - \frac{64}{3} + \frac{32}{5} \right) = 2\pi \cdot \frac{480 - 320 + 96}{15} = 2\pi \cdot \frac{256}{15} = \frac{512\pi}{15}.$$

Volumes with known cross-sections

- 7 The base of a solid is the region between $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.

Side length of each square $= 4 - x^2$.

$V = \int_{-2}^2 (4 - x^2)^2 dx = \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_{-2}^2 = 2 \left(32 - \frac{64}{3} + \frac{32}{5} \right) = 2 \cdot \frac{256}{15} = \frac{512}{15}$ (same integrand as the washer-about- $y = 4$ problem, but without the extra π since cross-sections are squares, not circles).

- 8 The base of a solid is a circle of radius 2 ($x^2 + y^2 = 4$). Cross-sections perpendicular to the x -axis are semicircles. Find the volume.

At each x , the base chord has length $2y = 2\sqrt{4 - x^2}$, which is the diameter of the semicircular cross-section, so its radius is $\sqrt{4 - x^2}$. Semicircle area $= \frac{1}{2}\pi(\sqrt{4 - x^2})^2 = \frac{\pi}{2}(4 - x^2)$.

$V = \int_{-2}^2 \frac{\pi}{2}(4 - x^2) dx = \frac{\pi}{2} \left[4x - \frac{x^3}{3} \right]_{-2}^2 = \frac{\pi}{2} \cdot 2 \left(8 - \frac{8}{3} \right) = \pi \cdot \frac{16}{3} = \frac{16\pi}{3}$.

- 9 Set up (but do not evaluate) the integral for the volume of the solid whose base is the region bounded by $y = \sin x$ and the x -axis on $[0, \pi]$, with cross-sections perpendicular to the x -axis that are equilateral triangles.

Each equilateral triangle has side length $s = \sin x$, so its area is $\frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}\sin^2 x$. $V = \int_0^\pi \frac{\sqrt{3}}{4}\sin^2 x dx$.

Volume by revolving about the y -axis (washer method)

- 10 Find the volume formed by revolving the region between $x = y^2$ and $x = 2y$ about the y -axis.

Intersections: $y^2 = 2y \Rightarrow y(y - 2) = 0$; $y = 0, 2$. On $[0, 2]$, outer radius $R = 2y$, inner radius $r = y^2$.

$V = \pi \int_0^2 [(2y)^2 - (y^2)^2] dy = \pi \int_0^2 (4y^2 - y^4) dy = \pi \left[\frac{4y^3}{3} - \frac{y^5}{5} \right]_0^2 = \pi \left(\frac{32}{3} - \frac{32}{5} \right) = \pi \cdot \frac{160 - 96}{15} = \frac{64\pi}{15}$.

Visualizing a solid of revolution's generating region

- 11 The graph shows $y = \sqrt{x}$ on $[0, 4]$ from the first disc-method problem. Sketching this region mentally revolved about the x -axis produces a solid — use the volume 8π found earlier to find its average cross-sectional area over the interval, and compare that to the cross-sectional area at the midpoint $x = 2$.

Average cross-sectional area $= \frac{V}{4} = \frac{8\pi}{4} = 2\pi$ (dividing volume by the interval's length). At $x = 2$, the cross-section is a disc of radius $\sqrt{2}$, area $= \pi(\sqrt{2})^2 = 2\pi$ — matching the average exactly, since the cross-sectional area πx is linear in x , so its average equals its midpoint value.

A capstone washer-method problem

- 12 Find the volume formed by revolving the region between $y = 4 - x^2$ and $y = x^2$ about the x -axis (using the region where $4 - x^2 \geq x^2$).

Intersections: $4 - x^2 = x^2 \Rightarrow x^2 = 2 \Rightarrow x = \sqrt{2} \text{ or } -\sqrt{2}$. Outer radius $R = 4 - x^2$, inner radius $r = x^2$.

$$V = \pi \int_{-\sqrt{2}}^{\sqrt{2}} [(4 - x^2)^2 - (x^2)^2] dx = \pi \int_{-\sqrt{2}}^{\sqrt{2}} (16 - 8x^2) dx = \pi \left[16x - \frac{8x^3}{3} \right]_{-\sqrt{2}}^{\sqrt{2}} = 2\pi \left(16\sqrt{2} - \frac{16\sqrt{2}}{3} \right) = 2\pi \cdot \frac{32\sqrt{2}}{3} = \frac{64\sqrt{2}\pi}{3}$$

Volumes with rectangular cross-sections

- 13 The base of a solid is the region bounded by $y = x^2$ and $y = 1$. Cross-sections perpendicular to the x -axis are rectangles with height twice the base. Find the volume.

At each $x \in [-1, 1]$, the base width is $1 - x^2$, so the rectangle's area is $2(1 - x^2)^2$.

$$V = \int_{-1}^1 2(1 - x^2)^2 dx = 2 \int_{-1}^1 (1 - 2x^2 + x^4) dx = 2 \left[x - \frac{2x^3}{3} + \frac{x^5}{5} \right]_{-1}^1 = 2 \cdot 2 \left(1 - \frac{2}{3} + \frac{1}{5} \right) = 4 \cdot \frac{15 - 10 + 3}{15} = 4 \cdot \frac{8}{15} = \frac{32}{15}$$

A capstone volume application

- 14 A bowl is formed by revolving $y = \frac{x^2}{4}$ (for $0 \leq x \leq 4$, where x is horizontal distance and y is depth from the rim) about the y -axis. Set up (do not evaluate) the disc-method integral for the volume of the bowl in terms of y , using the fact that $x^2 = 4y$.

$$V = \pi \int_0^4 x^2 dy = \pi \int_0^4 4y dy \text{ (using } y \text{ as the integration variable and } x^2 = 4y \text{ from solving the original equation for } x^2 \text{).}$$

- 15 Find the volume formed by revolving $y = e^x$ about the x -axis on $[0, 1]$.

$$V = \pi \int_0^1 (e^x)^2 dx = \pi \int_0^1 e^{2x} dx = \pi \left[\frac{1}{2} e^{2x} \right]_0^1 = \frac{\pi}{2} (e^2 - 1).$$

- 16 A vase is designed by revolving $y = \frac{1}{2}x + 1$ about the x -axis on $[0, 6]$, where x is the height (inches) and y is the radius. Find the volume of the vase.

$$V = \pi \int_0^6 \left(\frac{x}{2} + 1 \right)^2 dx = \pi \int_0^6 \left(\frac{x^2}{4} + x + 1 \right) dx = \pi \left[\frac{x^3}{12} + \frac{x^2}{2} + x \right]_0^6 = \pi (18 + 18 + 6) = 42\pi \text{ in}^3.$$

- 17 For the region between $y = x$ and $y = x^3$ on $[0, 1]$ (where $y = x \geq y = x^3$), complete each step for the solid formed by revolving this region about the x -axis:

a Identify the outer radius R and inner radius r . Outer radius $R = x$ (the larger function on $[0, 1]$), inner radius $r = x^3$.

b Set up the washer-method integral. $V = \pi \int_0^1 [x^2 - x^6] dx$.

c Evaluate the integral. $V = \pi \left[\frac{x^3}{3} - \frac{x^7}{7} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{7} \right) = \frac{4\pi}{21}$.

Multiple choice — easy 15

18 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 5]$.

a. 626π

b. 125π

c. 625π ✓

d. $\frac{3125\pi}{6}$

$$V = \pi \int_0^5 (x^2)^2 dx = \pi \int_0^5 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^5 = \pi \cdot \frac{3125}{5} = 625\pi.$$

19 Find the volume formed by revolving $y = x^3$ about the x -axis on $[0, 4]$.

a. 2048π

b. $\frac{16391\pi}{7}$

c. $\frac{16384\pi}{7}$ ✓

d. $\frac{4096\pi}{7}$

$$V = \pi \int_0^4 (x^3)^2 dx = \pi \int_0^4 x^6 dx = \pi \left[\frac{x^7}{7} \right]_0^4 = \pi \cdot \frac{16384}{7} = \frac{16384\pi}{7}.$$

20 Find the volume formed by revolving $y = x^3$ about the x -axis on $[0, 6]$.

a. $\frac{46656\pi}{7}$

b. $\frac{279943\pi}{7}$

c. 34992π

d. $\frac{279936\pi}{7}$ ✓

$$V = \pi \int_0^6 (x^3)^2 dx = \pi \int_0^6 x^6 dx = \pi \left[\frac{x^7}{7} \right]_0^6 = \pi \cdot \frac{279936}{7} = \frac{279936\pi}{7}.$$

21 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 6]$.

a. $\frac{1296\pi}{5}$

b. $\frac{7781\pi}{5}$

c. 1296π

d. $\frac{7776\pi}{5}$ ✓

$$V = \pi \int_0^6 (x^2)^2 dx = \pi \int_0^6 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^6 = \pi \cdot \frac{7776}{5} = \frac{7776\pi}{5}.$$

22 Find the volume formed by revolving $y = x^3$ about the x -axis on $[0, 3]$.

a. $\frac{2187\pi}{8}$

b. $\frac{729\pi}{7}$

c. $\frac{2187\pi}{7}$ ✓

d. $\frac{2194\pi}{7}$

$$V = \pi \int_0^3 (x^3)^2 dx = \pi \int_0^3 x^6 dx = \pi \left[\frac{x^7}{7} \right]_0^3 = \pi \cdot \frac{2187}{7} = \frac{2187\pi}{7}.$$

23 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 5]$.

a. $\frac{128\pi}{3}$

b. $\frac{125\pi}{3}$ ✓

c. $\frac{25\pi}{3}$

d. $\frac{125\pi}{4}$

$$V = \pi \int_0^5 (x)^2 dx = \pi \int_0^5 x^2 dx = \pi \left[\frac{x^3}{3} \right]_0^5 = \pi \cdot \frac{125}{3} = \frac{125\pi}{3}.$$

24 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 3]$.

a. $\frac{248\pi}{5}$

b. $\frac{81\pi}{2}$

c. $\frac{81\pi}{5}$

d. $\frac{243\pi}{5}$ ✓

$$V = \pi \int_0^3 (x^2)^2 dx = \pi \int_0^3 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^3 = \pi \cdot \frac{243}{5} = \frac{243\pi}{5}.$$

25 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 2]$.

a. $\frac{32\pi}{5}$ ✓

b. $\frac{16\pi}{3}$

c. $\frac{16\pi}{5}$

d. $\frac{37\pi}{5}$

$$V = \pi \int_0^2 (x^2)^2 dx = \pi \int_0^2 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^2 = \pi \cdot \frac{32}{5} = \frac{32\pi}{5}.$$

26 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 2]$.

a. 2π

b. $\frac{4\pi}{3}$

c. $\frac{8\pi}{3}$ ✓

d. $\frac{11\pi}{3}$

$$V = \pi \int_0^2 (x)^2 dx = \pi \int_0^2 x^2 dx = \pi \left[\frac{x^3}{3} \right]_0^2 = \pi \cdot \frac{8}{3} = \frac{8\pi}{3}.$$

27 Find the volume formed by revolving $y = x^3$ about the x -axis on $[0, 2]$.

a. $\frac{128\pi}{7}$ ✓

b. $\frac{135\pi}{7}$

c. $\frac{64\pi}{7}$

d. 16π

$$V = \pi \int_0^2 (x^3)^2 dx = \pi \int_0^2 x^6 dx = \pi \left[\frac{x^7}{7} \right]_0^2 = \pi \cdot \frac{128}{7} = \frac{128\pi}{7}.$$

28 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 6]$.

a. 54π

b. 72π ✓

c. 73π

d. 12π

$$V = \pi \int_0^6 (x)^2 dx = \pi \int_0^6 x^2 dx = \pi \left[\frac{x^3}{3} \right]_0^6 = \pi \cdot \frac{216}{3} = 72\pi.$$

29 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 1]$.

a. $\frac{\pi}{3}$ ✓

b. $\frac{\pi}{4}$

c. $\frac{5\pi}{6}$

d. $\frac{4\pi}{3}$

$$V = \pi \int_0^1 (x)^2 dx = \pi \int_0^1 x^2 dx = \pi \left[\frac{x^3}{3} \right]_0^1 = \pi \cdot \frac{1}{3} = \frac{\pi}{3}.$$

30 Find the volume formed by revolving $y = x^2$ about the x -axis on $[0, 1]$.

a. $\frac{6\pi}{5}$

b. $\frac{\pi}{5}$ ✓

c. $\frac{\pi}{6}$

d. $\frac{7\pi}{10}$

$$V = \pi \int_0^1 (x^2)^2 dx = \pi \int_0^1 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^1 = \pi \cdot \frac{1}{5} = \frac{\pi}{5}.$$

31 Find the volume formed by revolving $y = x^4$ about the x -axis on $[0, 2]$.

a. $\frac{512\pi}{9}$ ✓

b. $\frac{521\pi}{9}$

c. $\frac{256\pi}{5}$

d. $\frac{256\pi}{9}$

$$V = \pi \int_0^2 (x^4)^2 dx = \pi \int_0^2 x^8 dx = \pi \left[\frac{x^9}{9} \right]_0^2 = \pi \cdot \frac{512}{9} = \frac{512\pi}{9}.$$

32 Find the volume formed by revolving $y = x$ about the x -axis on $[0, 4]$.

- a. $\frac{16\pi}{3}$
- b. $\frac{67\pi}{3}$
- c. 16π
- d. $\frac{64\pi}{3}$ ✓

$$V = \pi \int_0^4 (x)^2 dx = \pi \int_0^4 x^2 dx = \pi \left[\frac{x^3}{3} \right]_0^4 = \pi \cdot \frac{64}{3} = \frac{64\pi}{3}.$$

Multiple choice — medium 15

33 Find the volume formed by revolving the region between $y = x^4$ and $y = x^8$ (on $[0, 1]$) about the x -axis.

- a. $\frac{8\pi}{153}$ ✓
- b. $\frac{4\pi}{45}$
- c. $-\frac{8\pi}{153}$
- d. $\frac{26\pi}{153}$

$$\text{Outer radius } R = x^4, \text{ inner radius } r = x^8. V = \pi \int_0^1 [x^8 - x^{16}] dx = \pi \left[\frac{x^9}{9} - \frac{x^{17}}{17} \right]_0^1 = \pi \left(\frac{1}{9} - \frac{1}{17} \right) = \frac{8\pi}{153}.$$

34 Find the volume formed by revolving the region between $y = x^4$ and $y = x^6$ (on $[0, 1]$) about the x -axis.

- a. $-\frac{4\pi}{117}$
- b. $\frac{4\pi}{117}$ ✓
- c. $\frac{2\pi}{35}$
- d. $\frac{22\pi}{117}$

$$\text{Outer radius } R = x^4, \text{ inner radius } r = x^6. V = \pi \int_0^1 [x^8 - x^{12}] dx = \pi \left[\frac{x^9}{9} - \frac{x^{13}}{13} \right]_0^1 = \pi \left(\frac{1}{9} - \frac{1}{13} \right) = \frac{4\pi}{117}.$$

35 Find the volume formed by revolving the region between $y = x$ and $y = x^4$ (on $[0, 1]$) about the x -axis.

- a. $\frac{3\pi}{10}$
- b. $\frac{2\pi}{9}$ ✓
- c. $-\frac{2\pi}{9}$
- d. $\frac{4\pi}{9}$

Outer radius $R = x$, inner radius $r = x^4$. $V = \pi \int_0^1 [x^2 - x^8] dx = \pi \left[\frac{x^3}{3} - \frac{x^9}{9} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{9} \right) = \frac{2\pi}{9}$.

36 Find the volume formed by revolving the region between $y = x^3$ and $y = x^7$ (on $[0, 1]$) about the x -axis.

- a. $\frac{8\pi}{105}$ ✓
- b. $\frac{22\pi}{105}$
- c. $\frac{\pi}{8}$
- d. $-\frac{8\pi}{105}$

Outer radius $R = x^3$, inner radius $r = x^7$. $V = \pi \int_0^1 [x^6 - x^{14}] dx = \pi \left[\frac{x^7}{7} - \frac{x^{15}}{15} \right]_0^1 = \pi \left(\frac{1}{7} - \frac{1}{15} \right) = \frac{8\pi}{105}$.

37 Find the volume formed by revolving the region between $y = x$ and $y = x^6$ (on $[0, 1]$) about the x -axis.

- a. $\frac{10\pi}{39}$ ✓
- b. $\frac{16\pi}{39}$
- c. $-\frac{10\pi}{39}$
- d. $\frac{5\pi}{14}$

Outer radius $R = x$, inner radius $r = x^6$. $V = \pi \int_0^1 [x^2 - x^{12}] dx = \pi \left[\frac{x^3}{3} - \frac{x^{13}}{13} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{13} \right) = \frac{10\pi}{39}$.

38 Find the volume formed by revolving the region between $y = x^3$ and $y = x^9$ (on $[0, 1]$) about the x -axis.

- a. $\frac{3\pi}{20}$
- b. $-\frac{12\pi}{133}$
- c. $\frac{12\pi}{133}$ ✓
- d. $\frac{26\pi}{133}$

Outer radius $R = x^3$, inner radius $r = x^9$. $V = \pi \int_0^1 [x^6 - x^{18}] dx = \pi \left[\frac{x^7}{7} - \frac{x^{19}}{19} \right]_0^1 = \pi \left(\frac{1}{7} - \frac{1}{19} \right) = \frac{12\pi}{133}$.

39 Find the volume formed by revolving the region between $y = x^2$ and $y = x^4$ (on $[0, 1]$) about the x -axis.

a. $\frac{4\pi}{45}$ ✓

b. $\frac{14\pi}{45}$

c. $\frac{2\pi}{15}$

d. $-\frac{4\pi}{45}$

Outer radius $R = x^2$, inner radius $r = x^4$. $V = \pi \int_0^1 [x^4 - x^8] dx = \pi \left[\frac{x^5}{5} - \frac{x^9}{9} \right]_0^1 = \pi \left(\frac{1}{5} - \frac{1}{9} \right) = \frac{4\pi}{45}$.

40 Find the volume formed by revolving the region between $y = x^2$ and $y = x^6$ (on $[0, 1]$) about the x -axis.

a. $\frac{8\pi}{65}$ ✓

b. $-\frac{8\pi}{65}$

c. $\frac{4\pi}{21}$

d. $\frac{18\pi}{65}$

Outer radius $R = x^2$, inner radius $r = x^6$. $V = \pi \int_0^1 [x^4 - x^{12}] dx = \pi \left[\frac{x^5}{5} - \frac{x^{13}}{13} \right]_0^1 = \pi \left(\frac{1}{5} - \frac{1}{13} \right) = \frac{8\pi}{65}$.

41 Find the volume formed by revolving the region between $y = x^3$ and $y = x^6$ (on $[0, 1]$) about the x -axis.

a. $\frac{6\pi}{91}$ ✓

b. $\frac{20\pi}{91}$

c. $\frac{3\pi}{28}$

d. $-\frac{6\pi}{91}$

Outer radius $R = x^3$, inner radius $r = x^6$. $V = \pi \int_0^1 [x^6 - x^{12}] dx = \pi \left[\frac{x^7}{7} - \frac{x^{13}}{13} \right]_0^1 = \pi \left(\frac{1}{7} - \frac{1}{13} \right) = \frac{6\pi}{91}$.

42 Find the volume formed by revolving the region between $y = x^4$ and $y = x^7$ (on $[0, 1]$) about the x -axis.

a. $\frac{2\pi}{45}$ ✓

b. $-\frac{2\pi}{45}$

c. $\frac{3\pi}{40}$

d. $\frac{8\pi}{45}$

Outer radius $R = x^4$, inner radius $r = x^7$. $V = \pi \int_0^1 [x^8 - x^{14}] dx = \pi \left[\frac{x^9}{9} - \frac{x^{15}}{15} \right]_0^1 = \pi \left(\frac{1}{9} - \frac{1}{15} \right) = \frac{2\pi}{45}$.

43 Find the volume formed by revolving the region between $y = x^3$ and $y = x^8$ (on $[0, 1]$) about the x -axis.

a. $\frac{10\pi}{119}$ ✓

b. $\frac{5\pi}{36}$

c. $-\frac{10\pi}{119}$

d. $\frac{24\pi}{119}$

Outer radius $R = x^3$, inner radius $r = x^8$. $V = \pi \int_0^1 [x^6 - x^{16}] dx = \pi \left[\frac{x^7}{7} - \frac{x^{17}}{17} \right]_0^1 = \pi \left(\frac{1}{7} - \frac{1}{17} \right) = \frac{10\pi}{119}$.

44 Find the volume formed by revolving the region between $y = x^4$ and $y = x^9$ (on $[0, 1]$) about the x -axis.

a. $\frac{\pi}{10}$

b. $-\frac{10\pi}{171}$

c. $\frac{28\pi}{171}$

d. $\frac{10\pi}{171}$ ✓

Outer radius $R = x^4$, inner radius $r = x^9$. $V = \pi \int_0^1 [x^8 - x^{18}] dx = \pi \left[\frac{x^9}{9} - \frac{x^{19}}{19} \right]_0^1 = \pi \left(\frac{1}{9} - \frac{1}{19} \right) = \frac{10\pi}{171}$.

45 Find the volume formed by revolving the region between $y = x^2$ and $y = x^8$ (on $[0, 1]$) about the x -axis.

a. $\frac{2\pi}{9}$

b. $\frac{22\pi}{85}$

c. $-\frac{12\pi}{85}$

d. $\frac{12\pi}{85}$ ✓

Outer radius $R = x^2$, inner radius $r = x^8$. $V = \pi \int_0^1 [x^4 - x^{16}] dx = \pi \left[\frac{x^5}{5} - \frac{x^{17}}{17} \right]_0^1 = \pi \left(\frac{1}{5} - \frac{1}{17} \right) = \frac{12\pi}{85}$.

46 Find the volume formed by revolving the region between $y = x^3$ and $y = x^4$ (on $[0, 1]$) about the x -axis.

- a. $\frac{\pi}{20}$
- b. $\frac{2\pi}{63}$ ✓
- c. $\frac{16\pi}{63}$
- d. $-\frac{2\pi}{63}$

Outer radius $R = x^3$, inner radius $r = x^4$. $V = \pi \int_0^1 [x^6 - x^8] dx = \pi \left[\frac{x^7}{7} - \frac{x^9}{9} \right]_0^1 = \pi \left(\frac{1}{7} - \frac{1}{9} \right) = \frac{2\pi}{63}$.

47 Find the volume formed by revolving the region between $y = x$ and $y = x^2$ (on $[0, 1]$) about the x -axis.

- a. $\frac{\pi}{6}$
- b. $\frac{2\pi}{15}$ ✓
- c. $\frac{8\pi}{15}$
- d. $-\frac{2\pi}{15}$

Outer radius $R = x$, inner radius $r = x^2$. $V = \pi \int_0^1 [x^2 - x^4] dx = \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}$.

Multiple choice — hard 15

48 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 16$ about the line $y = 16$.

- a. $\frac{16384\pi}{15}$ ✓
- b. $\frac{8192\pi}{15}$
- c. $\frac{4096\pi}{15}$
- d. $\frac{16444\pi}{15}$

Intersections at $x = +4$. Radius₄ = $16 - x^2$.

$$V = \pi \int_{-4}^4 (16 - x^2)^2 dx = 2\pi \int_0^4 (16 - x^2)^2 dx = \frac{16\pi(4)^5}{15} = \frac{16384\pi}{15}.$$

49 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 9$ about the line $y = 9$.

a. $\frac{1311\pi}{5}$

b. $\frac{648\pi}{5}$

c. $\frac{1296\pi}{5}$ ✓

d. $\frac{432\pi}{5}$

Intersections at $x = \pm 3$. Radius = $9 - x^2$. $V = \pi \int_{-3}^3 (9 - x^2)^2 dx = 2\pi \int_0^3 (9 - x^2)^2 dx = \frac{16\pi(3)^5}{15} = \frac{1296\pi}{5}$.

50 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 4$ about the line $y = 4$.

a. $\frac{256\pi}{15}$

b. $\frac{542\pi}{15}$

c. $\frac{1039\pi}{30}$

d. $\frac{512\pi}{15}$ ✓

Intersections at $x = \pm 2$. Radius = $4 - x^2$. $V = \pi \int_{-2}^2 (4 - x^2)^2 dx = 2\pi \int_0^2 (4 - x^2)^2 dx = \frac{16\pi(2)^5}{15} = \frac{512\pi}{15}$.

51 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 25$ about the line $y = 25$.

a. $\frac{5000\pi}{3}$

b. $\frac{10000\pi}{3}$ ✓

c. $\frac{10015\pi}{3}$

d. $\frac{2000\pi}{3}$

Intersections at $x = \pm 5$. Radius = $25 - x^2$.
 $V = \pi \int_{-5}^5 (25 - x^2)^2 dx = 2\pi \int_0^5 (25 - x^2)^2 dx = \frac{16\pi(5)^5}{15} = \frac{10000\pi}{3}$.

52 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 1$ about the line $y = 1$.

- a. $\frac{17\pi}{30}$
- b. $\frac{31\pi}{15}$
- c. $\frac{8\pi}{15}$
- d. $\frac{16\pi}{15}$ ✓

Intersections at $x = \pm 1$. Radius = $1 - x^2$. $V = \pi \int_{-1}^1 (1 - x^2)^2 dx = 2\pi \int_0^1 (1 - x^2)^2 dx = \frac{16\pi(1)^5}{15} = \frac{16\pi}{15}$.

53 The base of a solid is the region between $y = x^2$ and $y = 36$ on $[-6, 6]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.

- a. $\frac{6912}{5}$
- b. $\frac{41472}{5}$ ✓
- c. $\frac{20736}{5}$
- d. $\frac{41502}{5}$

Side length of each square = $36 - x^2$. $V = \int_{-6}^6 (36 - x^2)^2 dx = 2 \int_0^6 (36 - x^2)^2 dx = \frac{16(6)^5}{15} = \frac{41472}{5}$.

54 The base of a solid is the region between $y = x^2$ and $y = 25$ on $[-5, 5]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.

- a. $\frac{5000}{3}$
- b. $\frac{10000}{3}$ ✓
- c. $\frac{2000}{3}$
- d. $\frac{10015}{3}$

Side length of each square = $25 - x^2$. $V = \int_{-5}^5 (25 - x^2)^2 dx = 2 \int_0^5 (25 - x^2)^2 dx = \frac{16(5)^5}{15} = \frac{10000}{3}$.

- 55 The base of a solid is the region between $y = x^2$ and $y = 16$ on $[-4, 4]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.

- a. $\frac{8192}{15}$
 b. $\frac{4096}{15}$
 c. $\frac{16384}{15}$ ✓
 d. $\frac{16444}{15}$

Side length of each square $= 16 - x^2$. $V = \int_{-4}^4 (16 - x^2)^2 dx = 2 \int_0^4 (16 - x^2)^2 dx = \frac{16(4)^5}{15} = \frac{16384}{15}$.

- 56 The base of a solid is the region between $y = x^2$ and $y = 1$ on $[-1, 1]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.

- a. $\frac{8}{15}$
 b. $\frac{47}{30}$
 c. $\frac{16}{15}$ ✓
 d. $\frac{31}{15}$

Side length of each square $= 1 - x^2$. $V = \int_{-1}^1 (1 - x^2)^2 dx = 2 \int_0^1 (1 - x^2)^2 dx = \frac{16(1)^5}{15} = \frac{16}{15}$.

- 57 The base of a solid is the region between $y = x^2$ and $y = 9$ on $[-3, 3]$. Cross-sections perpendicular to the x -axis are squares. Find the volume.

- a. $\frac{1311}{5}$
 b. $\frac{648}{5}$
 c. $\frac{432}{5}$
 d. $\frac{1296}{5}$ ✓

Side length of each square $= 9 - x^2$. $V = \int_{-3}^3 (9 - x^2)^2 dx = 2 \int_0^3 (9 - x^2)^2 dx = \frac{16(3)^5}{15} = \frac{1296}{5}$.

- 58 The base of a solid is the region bounded by $y = x^2$ and $y = 9$ on $[-3, 3]$. Cross-sections perpendicular to the x -axis are rectangles with height 2 times the base. Find the volume.

a. $\frac{1296}{5}$

b. $\frac{2592}{5}$ ✓

c. $\frac{2607}{5}$

d. $\frac{5179}{10}$

At each x , the base width is $9 - x^2$, so the rectangle's area is $2(9 - x^2)^2$.
 $V = 2 \int_{-3}^3 (9 - x^2)^2 dx = 2 \cdot \frac{16(3)^5}{15} = \frac{2592}{5}$.

- 59 The base of a solid is the region bounded by $y = x^2$ and $y = 1$ on $[-1, 1]$. Cross-sections perpendicular to the x -axis are rectangles with height 2 times the base. Find the volume.

a. $\frac{49}{30}$

b. $\frac{32}{15}$ ✓

c. $\frac{16}{15}$

d. $\frac{47}{15}$

At each x , the base width is $1 - x^2$, so the rectangle's area is $2(1 - x^2)^2$.
 $V = 2 \int_{-1}^1 (1 - x^2)^2 dx = 2 \cdot \frac{16(1)^5}{15} = \frac{32}{15}$.

- 60 The base of a solid is the region bounded by $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x -axis are rectangles with height 2 times the base. Find the volume.

a. $\frac{2033}{30}$

b. $\frac{512}{15}$

c. $\frac{1054}{15}$

d. $\frac{1024}{15}$ ✓

At each x , the base width is $4 - x^2$, so the rectangle's area is $2(4 - x^2)^2$.
 $V = 2 \int_{-2}^2 (4 - x^2)^2 dx = 2 \cdot \frac{16(2)^5}{15} = \frac{1024}{15}$.

- 61 The base of a solid is the region bounded by $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x -axis are rectangles with height 4 times the base. Find the volume.

a. $\frac{2078}{15}$

b. $\frac{1024}{15}$

c. $\frac{2048}{15}$ ✓

d. $\frac{512}{15}$

At each x , the base width is $4 - x^2$, so the rectangle's area is $4(4 - x^2)^2$.

$$V = 4 \int_{-2}^2 (4 - x^2)^2 dx = 4 \cdot \frac{16(2)^5}{15} = \frac{2048}{15}.$$

- 62 The base of a solid is the region bounded by $y = x^2$ and $y = 9$ on $[-3, 3]$. Cross-sections perpendicular to the x -axis are rectangles with height 4 times the base. Find the volume.

a. $\frac{2592}{5}$

b. $\frac{5184}{5}$ ✓

c. $\frac{1296}{5}$

d. $\frac{5199}{5}$

At each x , the base width is $9 - x^2$, so the rectangle's area is $4(9 - x^2)^2$.

$$V = 4 \int_{-3}^3 (9 - x^2)^2 dx = 4 \cdot \frac{16(3)^5}{15} = \frac{5184}{5}.$$